

1.0 NAME OF THE VETERINARY MEDICINAL PRODUCT
Ketamina Biowet Pulawy 100mg/ml solution for injection

3.0 PHARMACEUTICAL FORM

Solution for injection.

Clear solution.

4.0 CLINICAL PARTICULARS

4.1 Target species

Dogs and Cats

4.2 Indications for use, specifying the target species

Ketamine is used for induction and/or maintenance of anaesthesia. It can be used alone in cats, after premedication and/or in combination with other anaesthetics in dogs and cats. The main indications are as follows:

Ketamine used alone or with premedication:

- Restraint, immobilisation.
 - Minor surgical procedure (superficial or of short duration).
- Ketamine used in combination, with or without premedication:
- Major surgical procedures (deep layers)
 - Emergency surgical procedures
 - Obstetric surgeries

4.3 Contraindications

Do not use in cases of hypersensitivity to the active substance or to any of the excipients.

Do not use in animals with:

- Intracranial hypertension
- Cardiac decompensation.
- Hepatic and/or renal failure.
- Glaucoma.
- Hyperthyroidism.
- Ocular surgery

4.5 Special precautions for use

Special precautions for use in animals

The animal's eyes remain open during anaesthesia. Precautions should be taken to avoid corneal drying. In dogs, use as a sole agent to induce anaesthesia is not advised. Premedication with other drugs is recommended to minimise possible side effects. Administration of an anticholinergic, such as atropine, may reduce the hypersalivation caused by ketamine, especially in cats.

Specific precautions to be taken by the person administering the veterinary medicinal product to animals

Persons with known hypersensitivity to ketamine should avoid contact with the veterinary medicinal product. Administer the product with caution, ketamine is an anaesthetic/analgesic agent.

In case of accidental self-injection, consult a doctor immediately and show the package leaflet or the label.

Wash hands thoroughly after use.

In case of accidental spillage on the skin, wash the area with soap and water.

In case of accidental eye contact, flush thoroughly with water and seek medical advice if signs or symptoms occur.

Warnings to physician: Do not leave patient unattended. There is no specific antidote for overexposure to ketamine hydrochloride. Maintain breathing and apply symptomatic maintenance treatment.

4.6 Adverse reactions (frequency and seriousness)

The main adverse reactions of ketamine observed in very rare cases include:

- Pain and local irritation at the injection site after intramuscular administration.
- Anaphylactic reactions.
- Signs that may be related to secondary pharmacological action.

The main signs related to secondary pharmacological action are:

In dogs, very rarely:

- General respiratory and cardiac signs: shock, tachycardia (which may cause cardiac arrest), tracheobronchial mucous secretions, tachypnoea, pulmonary oedema, respiratory arrest.
- Nervous signs: both nervous hyperexcitability (increased muscle tone, hyperaesthesia, convulsions) and signs due to prolonged anaesthesia (ataxia, catalepsy, salivation, balance disorders), especially in unmedicated dogs.
- Increased cerebral blood flow and cerebral oxygen consumption.
- Urinary disorders, such as haematuria.

In cats:

Cardiorespiratory disorders (tachycardia, cardiac arrest, apnoea and dyspnoea) and convulsions have been observed very rarely. Violent and disorientated recoveries are not uncommon. In a frequently occurring form of reaction in cats, catalepsy is followed by ataxia and increased motor activity, and in some cases there may be behavioural changes up to several hours after recovery. The incidence of arousal reactions on awakening is considerably reduced in the absence of tactile, auditory and visual stimuli, and if diazepam was administered as a premedication.

The frequency of adverse reactions should be classified according to the following groups:

- Very frequently (more than 1 animal per 10 animals treated exhibits adverse reactions)
- Frequently (more than 1 but less than 10 animals per 100 animals treated)
- Infrequently (more than 1 but less than 10 animals per 1,000 animals treated)
- Rarely (more than 1 but less than 10 animals per 10,000 animals treated)
- Very rarely (less than 1 animal per 10,000 animals treated, including isolated cases).

4.7 Use during pregnancy and lactation

Use is not recommended during the organogenic period (i.e. up to 4-5 weeks of gestation in dogs or 4-6 weeks in cats). No reproductive effects were observed in dams in late gestation. As ketamine crosses the placenta rapidly, transient depression may occur in neonates when ketamine is administered close to the time of delivery, so appropriate care should be given to neonates. Use only in accordance with the benefit/risk assessment carried out by the responsible veterinarian.

4.8 Interaction with other medicinal products and other forms of interaction

Do not use concomitantly with cholinesterase inhibitors (organophosphates). Special care should be taken when using ketamine-halothane combinations, as the half-life of ketamine is prolonged. Neuroleptoanalgesics, tranquilisers and chloramphenicol potentiate ketamine anaesthesia. Barbiturates and opioids may prolong the recovery period.

2.0 QUALITATIVE AND QUANTITATIVE COMPOSITION

Each ml contains:

Active substance:

Ketamine 100 mg
(equivalent to 115.34 mg ketamine hydrochloride)

Preservative:

Chlorobutanol Hemihydrate 3 mg

For the full list of excipients, see section 6.1.

4.4 Special warnings for each target species

It is important that both induction and recovery take place in a calm and quiet environment.

It is generally considered good anaesthetic practice to fast animals for approximately 12 hours before undergoing general anaesthesia.

4.9 Amounts to be administered and administration route

Routes of administration: intravenous or intramuscular.

Standard recommended dosage (when used alone, without premedication):

Intravenous route:

Administer 5-8 mg ketamine/kg (equivalent to 0.05 - 0.08 ml drug/kg).

Intramuscular route:

Administer 15-20 mg ketamine/kg (equivalent to 0.15 - 0.2 ml drug/kg).

Individual responses may vary. The recommended dosage therefore depends on the depth of anesthesia desired, the route of administration and the other drugs administered together. The intervals given are indicative. When used with premedication and/or in conjunction with other anesthetic drugs, the doses to be administered are lower. Premedication should be administered 15-20 minutes prior to induction. If necessary, anesthesia may be prolonged with maintenance doses. Maintenance dosage according to the need:

- 1st injection: $\frac{1}{2}$ of initial dose

- subsequent injections: $\frac{1}{4}$ of the initial dose.

However the actual dose will be based on the individual patient's response.

4.10 Overdose (symptoms, emergency procedures, antidotes), if necessary

Overdosage should be avoided at all times. At a dose higher than recommended, the risk of adverse reactions will increase and recovery time will be longer. The main adverse reactions related to overdosage are cardio-respiratory (cardiac arrest) and nervous disorders (convulsions, ataxia, mydriasis). High doses (25-40 mg/kg) in the cat increase the risk of seizures. In order not to overdose, it is important to take into account the co-administration of other sedative substances in calculating the appropriate dose of ketamine to be administered.

No antidote has been identified. However, symptomatic therapy may be beneficial.

4.11 Withdrawal period(s)

Not applicable

5. PHARMACOLOGICAL PROPERTIES

Pharmacotherapeutic group: General anaesthetics.

ATC vet code: QN01AX03.

5.1 Pharmacodynamic properties

Ketamine is an analgesic-anaesthetic agent that acts as a non-competitive antagonist of N-methyl-D-aspartate (NMDA) receptors involved in the transmission and modulation of nociceptive information in the spinal cord. The nerve impulse is blocked at the level of the cerebral cortex with some activation of the subcortical area, resulting in dissociative anaesthesia that induces narcosis and superficial analgesia without inducing depression of muscle and bulbar tone, with some reflexes, such as swallowing, being maintained. Ketamine also interacts with opioid, muscarinic and monoaminergic receptors, with some calcium ion channels and with the catecholamine system, especially with dopaminergic receptors. At anaesthetic doses, ketamine induces bronchodilation (sympathomimetic effect), increases heart rate, arterial blood pressure and intraocular pressure, and produces hypersalivation.

5.2 Pharmacokinetic particulars

When administered intravenously to dogs, ketamine is rapidly distributed (with a mean distribution half-life value of 1.95 min), into heavily irrigated tissues such as the brain and then redistributed to less irrigated tissues, so that it has a high volume of distribution (55% of body weight for the central compartment). Ketamine is highly lipid soluble and does not have a high plasma protein binding capacity (approximately 50% in the dog), which facilitates its rapid transfer across the blood-brain barrier. It is also able to cross the placenta. Elimination half-life is approximately 1 hour). It is metabolised mainly in the liver to norketamine, part of which is oxidised to dehydronorketamine. Both metabolites are excreted mainly in the urine. Ketamine is rapidly absorbed when administered intramuscularly (tmax: 5-15 min). Ketamine is virtually no longer present in the body by approximately 7 hours after injection.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Chlorobutanol Hemihydrate

Water for injections

6.2 Major incompatibilities

Do not combine with barbiturates or diazepam in the same syringe due to chemical incompatibility.

Do not mix with any other veterinary medicinal product.

6.3 Shelf life

Shelf-life of the veterinary medicinal product as packaged for sale: 2 years. Shelf-life after first opening the immediate packaging: 28 days.

6.4 Special precautions for storage

Store below 25 Celcius (air conditioning). Protect from light. Do not Freeze.

6.5 Nature and composition of immediate packaging

50mL brown amber glass vial Type II with bromobutyl rubber stopper with silver aluminium caps. One such vial packed into an outer carton with a pack insert

6.6 Special precautions for the disposal of unused veterinary medicinal product or waste materials derived from the use of such products

Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.

7. PRODUCT REGISTRATION HOLDER

KM Vet Pharm Sdn Bhd,

NO.1, JLN INDUSTRI BATU CAVES 1/1,

TMN PERINDUSTRIAN BATU CAVES,

68100 BATU CAVES,

MALAYSIA.

8. MANUFACTURER

Biowet Pulawy Sp. Z o.o., Arciucha 2, 24-100 Putawy, Poland

9. DATE OF REVISION OF THE TEXT

August 2023